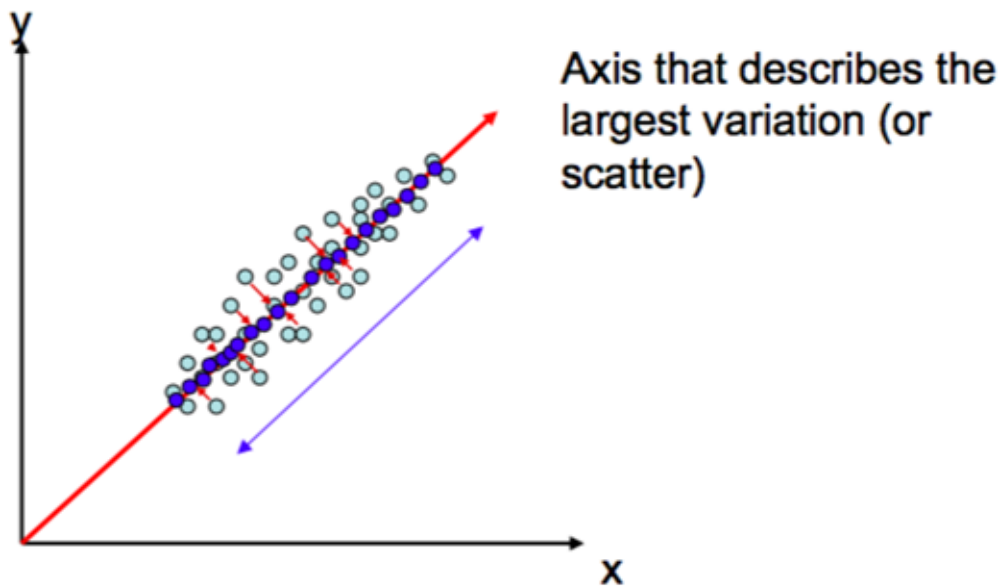
# DIMENSIONALITY REDUCTION AND VISUALIZATION

# What is PCA?

- We want to reduce the dimensionality but keep useful information
  - What is useful information? Variation
- We want to find a projection (a transformation) that describe maximum variation



# Formulation

- Maximize the variance after projection ie
  - argmax Var($\mathbf{w}^t\,\mathbf{x}$)
- Subject to w is a unit vector

- $\Sigma\mathbf{w} = \lambda\mathbf{w}$ <- eigenvector

# Selecting eigenvectors

- Remember the variance of projected data is

$$\omega^T \Sigma \omega. \quad (1)$$

- And our solution yielded

$$\Sigma\omega = \lambda\omega \quad (2)$$

- Plug (2) in (1) and we get

$$\begin{aligned} \text{projected variance} &= \omega^T \Sigma \omega = \omega^T \lambda\omega \\ &= \lambda\, \omega^T \omega \qquad (\text{remember } \|\omega\|=1) \\ &= \lambda \end{aligned}$$

# PCA

- The direction vector captures the variance corresponding to the eigenvalue
- So we want the higher eigenvalues
  - How many?

# Matrix rank

- A square dxd matrix has full rank (e.g. rank d) if the columns are linearly independent.
- The number of linearly independent columns is the rank of the matrix

- A covariance matrix of size dxd will have have at most N-1 rank where N is the number of training samples
  - 640x640 images = ~400000 dimensions
  - 1000 training images
  - The covariance matrix will be at most rank 999. The missing rank is because of the mean.

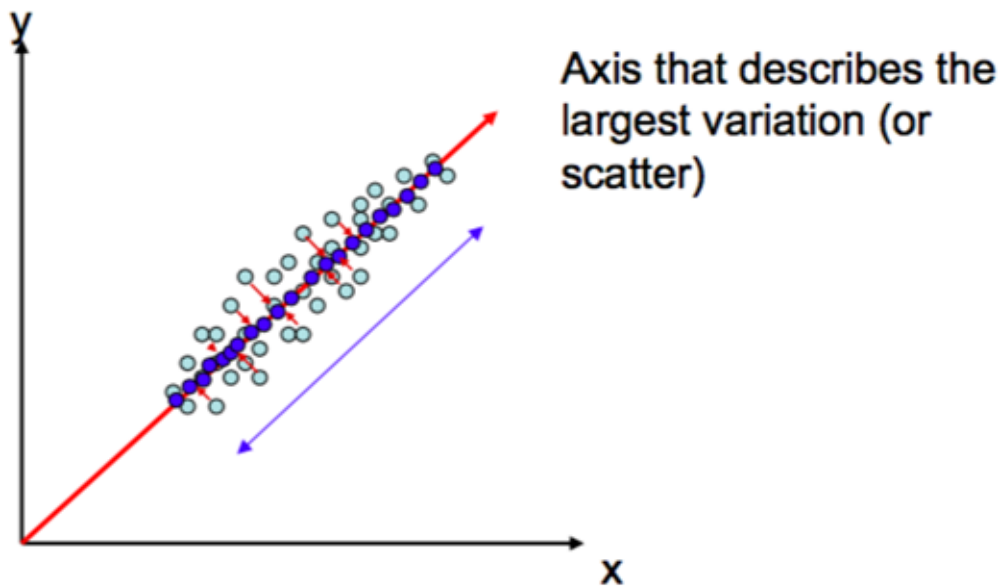
# PCA

- The direction vector captures the variance corresponding to the eigenvalue
- So we want the higher eigenvalues
- Take the eigenvalues with non-zero eigenvalues (at most N-1 non-zero eigenvalues)

# Eigenfaces

# What is PCA?

- We want to reduce the dimensionality but keep useful information
  - What is useful information? Variation
- We want to find a projection (a transformation) that describe maximum variation



# Means

- In PCA, we model variance. (Variation around the mean)
- In our projection we need to remove the mean

$$\mathbf{p} = \mathbf{V}^{\mathbf{T}}(\mathbf{x} - \mathbf{m})$$

- The mean is the mean of all your training data
- If we want to reconstruct the data we need to add back the mean

$$\mathbf{x} = \sum_{i=1}^{N} p_i \mathbf{v}_i + \mathbf{m} = p_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + p_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} + .. + p_n \begin{bmatrix} | \\ \mathbf{v}_n \\ | \end{bmatrix} + \mathbf{m} = \mathbf{V}\mathbf{p} + \mathbf{m}$$

# Basis decomposition

- Let's consider our projection $w_i$ which is the eigenvectors to be a basis vector $v_i$
- We can represent any vector as a sum of basis vectors as follows:

$$\mathbf{x} = \sum_{i=1}^{N} p_i \mathbf{v}_i = p_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + p_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} + .. + p_n \begin{bmatrix} | \\ \mathbf{v}_n \\ | \end{bmatrix} = \mathbf{V}\mathbf{p}$$

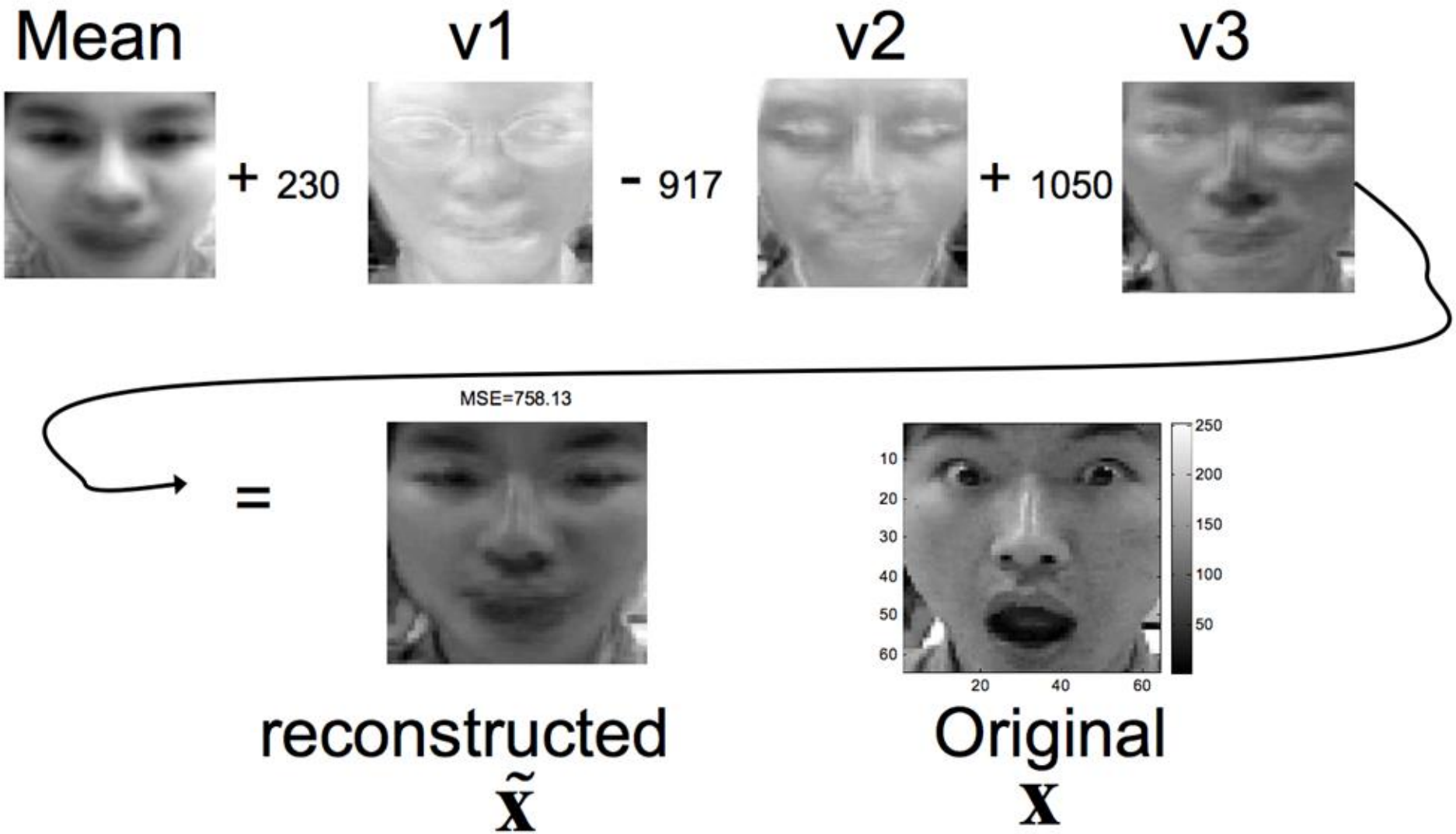
# Finding the weights

$$\mathbf{x} = \sum_{i=1}^{N} p_i \mathbf{v}_i = p_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + p_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} + .. + p_n \begin{bmatrix} | \\ \mathbf{v}_n \\ | \end{bmatrix} = \mathbf{V}\mathbf{p}$$

- If $v_i$ are orthogonal, the projection of x onto $v_i$ gives $p_i$

$$\mathbf{V}^{\mathrm{T}}\mathbf{x} = \begin{bmatrix} - & \mathbf{v}_1 & - \\ - & \mathbf{v}_2 & - \\ - & \mathbf{v}_3 & - \end{bmatrix} \begin{bmatrix} | \\ \mathbf{x} \\ | \end{bmatrix} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

# Reconstruction with eigenfaces



# Practical issues

- If your data has different magnitudes in different dimensions, normalize each dimension before PCA

- If we have 640x640 images = ~400000 dimensions.
- What is the size of the covariance matrix?
- 400000 x 400000

# Practical issues

- You have N training examples.
- For the case where N << 400000, we only have N-1 eigenvalues we care about anyway

# Gram Matrix

$$\Sigma = E(x - \mu)(x - \mu)^T = XX^T$$

Covariance matrix is the outer-product of the input matrix

Must solve $\Sigma v = \lambda v$

$$XX^T v = \lambda v \quad \text{(pre-mult by } X^T\text{)} \quad (1)$$

$$X^T X X^T v = \lambda X^T v \quad (v' = X^T v) \quad (2)$$

Solve eigenvalue problem $X^T X v' = \lambda v'$

- $X^T X$ is a gram of inner-product matrix. Its size is NxN where N is the number of data samples.

# But how to get v from v'?

- From previous slide, equation (1) and (2)
  - $XX^Tv = \lambda v$ (1)
  - $v' = X^Tv$ (2)
- Substitute (2) into (1)
  - $Xv' = \lambda v$

- Thus, $v = Xv'$. We don't care about the scaling term because we will always scale the eigenvector so that it is orthonormal i.e. $||v|| = 1$.

# How many eigenvectors?

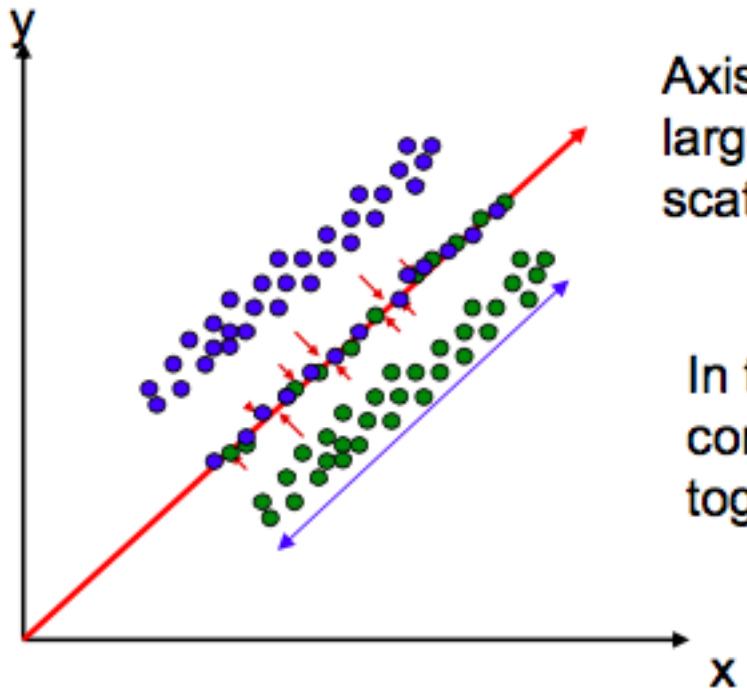
- Select based on amount of variance explained
  - Sum of eigenvalues exceeds some percent of total
- Reconstruction error

$$\mathbf{x} = \sum_{i=1}^{N} p_i \mathbf{v}_i = p_1 \begin{bmatrix} | \\ \mathbf{v}_1 \\ | \end{bmatrix} + p_2 \begin{bmatrix} | \\ \mathbf{v}_2 \\ | \end{bmatrix} + .. + p_n \begin{bmatrix} | \\ \mathbf{v}_n \\ | \end{bmatrix} = \mathbf{V}\mathbf{p}$$

  - Select enough v so that the difference between original x and reconstructed x is small

# PCA for classification

- PCA does not cares about the class labels

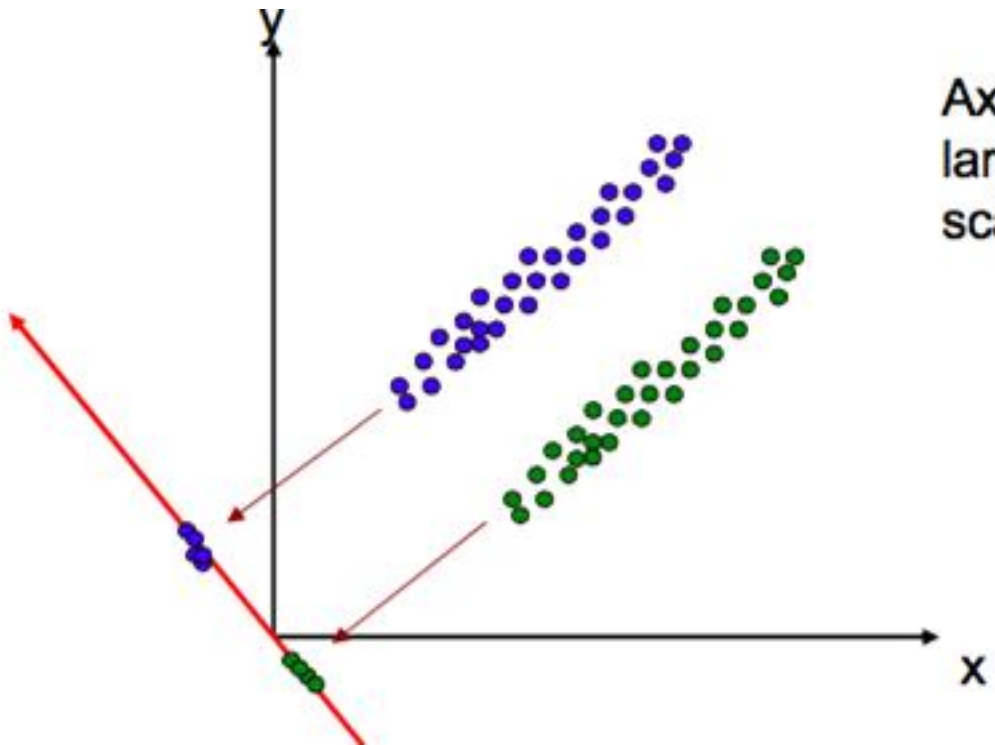



Axis that describes the largest variation (or scatter)…

In this case the projection vector completely smears the two classes together, making them in-separable

# What is LDA

- Find the projections that separate the classes.
- Assumes unimodal Gaussian model for each class
  - Maximize the distance between the means and minimize the variance of each class -> best classification performance




Axis that describes the largest variation (or scatter)

# Simple 2 class case

- We want to maximize the distance between the projected means:

  e.g. maximize $|(\tilde{\mu}_1 - \tilde{\mu}_2)|^2$

Where $\tilde{\mu}_1$ is the projected mean $\boldsymbol{\mu}_1$ of class onto LDA direction vector $\mathbf{w}$, i.e.

$$\tilde{\mu}_1 = \mathbf{w}^{\mathrm{T}}\boldsymbol{\mu}_1$$

and for class 2: $\tilde{\mu}_2 = \mathbf{w}^{\mathrm{T}}\boldsymbol{\mu}_2$ thus

$$|(\tilde{\mu}_1 - \tilde{\mu}_2)|^2 = |(\mathbf{w}^{\mathrm{T}}\boldsymbol{\mu}_1 - \mathbf{w}^{\mathrm{T}}\boldsymbol{\mu}_2)|^2$$

$$= \mathbf{w}^{\mathrm{T}}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\ (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^{\mathrm{T}}\mathbf{w}$$

$$= \mathbf{w}^{\mathrm{T}}\mathbf{S}_{\mathbf{B}}\mathbf{w}$$

# Between class scatter matrix $S_B$

$$(\tilde{\mu}_1 - \tilde{\mu}_2)^2 = (\mathbf{w}^T\boldsymbol{\mu}_1 - \mathbf{w}^T\boldsymbol{\mu}_2)^2$$

$$= \mathbf{w}^T(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)\ (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T\mathbf{w}$$

$$= \mathbf{w}^T\mathbf{S}_B\mathbf{w}$$

We want to maximize $w^TS_Bw$ where $S_B$ is the between class scatter matrix defined as:

$$\mathbf{S}_B = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T$$

# We also want to minimize within class scatter

- The variance or scatter of each class. We also want to minimize them.

$$\tilde{s}_1^{\,2} = \sum_{i=1}^{N_1} (\tilde{x}_i - \tilde{\mu}_1)^2$$

Minimize the total scatter $\tilde{s}_1^{\,2} + \tilde{s}_2^{\,2}$

# Within class scatter

- Lets expand on scatter $s_1, s_2$.

$$\tilde{s}_1^{\,2} = \sum_{i=1}^{N_1} (\tilde{x}_i - \tilde{\mu}_1)^2$$

$$= \sum_{i=1}^{N_1} (\mathbf{w}^{\mathrm{T}}\mathbf{x}_i - \mathbf{w}^{\mathrm{T}}\boldsymbol{\mu}_1)^2$$

$$= \sum_{i=1}^{N_1} \mathbf{w}^{\mathrm{T}}(\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^{\mathrm{T}}\mathbf{w}$$

$$= \mathbf{w}^{\mathrm{T}}\mathbf{S}_1\mathbf{w}$$

# Total within class scatter

- We want to minimize $$\tilde{s}_1^{\,2} + \tilde{s}_2^{\,2}$$

- This is the same as $\mathbf{w}^T\mathbf{S}_\mathbf{w}\mathbf{w}$

$$\mathbf{S}_\mathbf{w} = \sum_{i=1}^{C}\sum_{j=1}^{Ni} (\mathbf{x}_j - \boldsymbol{\mu}_i)(\mathbf{x}_j - \boldsymbol{\mu}_i)^\mathrm{T}$$

C number of classes, Ni number of images from class i

# Fisher Linear Discriminant Criterion

- We want to maximize between class scatter
- We want to minimize within class scatter

- We have an objective function as a ratio so we can achieve both!

$$J(\mathbf{w}) = \frac{|(\tilde{\mu}_1 - \tilde{\mu}_2)|^2}{\tilde{s}_1^{\,2} + \tilde{s}_2^{\,2}}$$

$$J(\mathbf{w}) = \frac{\mathbf{w}^{\mathrm{T}}\mathbf{S}_{\mathrm{B}}\mathbf{w}}{\mathbf{w}^{\mathrm{T}}\mathbf{S}_{\mathrm{W}}\mathbf{w}}$$

# LDA solution

- If you do calculus

$$\mathbf{S}_B \mathbf{w} = \lambda \mathbf{S}_W \mathbf{w}$$

$$\mathbf{S}_W^{-1} \mathbf{S}_B \mathbf{w} = \lambda \mathbf{w}$$

If $S_w$ is non-singular and invertible.

- Generalized eigenvalue problem. The number of solutions is min(rank$S_B$, rank$S_W$) = C-1 or N-C
- For 2 class this simplifies to
  - Note this is only one projection direction

$$\mathbf{w} = \mathbf{S}_W^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

# LDA+PCA

- Original features dimension L
- $S_B$ $S_W$ are LxL matrixes
- rank$S_B$ = C-1, rank$S_W$ = N-C
- First do PCA to reduce dimension
- Then do LDA to maximize classification ability
- How many dimensions to PCA?
  - Do PCA to keep N-C eigenvectors -> Makes $S_w$ full rank and invertible
  - Then, do LDA and compute C-1 projections in this N-C subspace
- PCA+LDA = Fisher projection

# Random projection

- Original d-dimensional data is project to k-dimensional subspace
- Using a random k x d matrix R with unit norm columns
  - Johnson-Lindenstrauss lemma: If points in a vector space are projected onto a randomly selected subspace of suitably high dimension, then the distances between the points are approximately preserved
- Elements of R are usually selected from Gaussians.
  - Generally any zero mean unit variance distribution would satisfy Johnson-Lindenstrauss lemma.

# Random projection notes

- R is not generally orthogonal.
  - But in a substantially large subspace, random vectors might be close to orthogonal.
- Looks weird but works…
- Can be used to initialized GMMs by finding good initializations in the projected space first

- Robust to outliers in the data, because it uses no data
- Faster to train than PCA if data has high dimension
  - $O(k^2 \times n + k^3)$ (covariance computation + eigen depcomposition) where k- features, n – samples
- See https://medium.com/data-science-in-your-pocket/random-projection-for-dimension-reduction-27d2ec7d40cd for usage guides

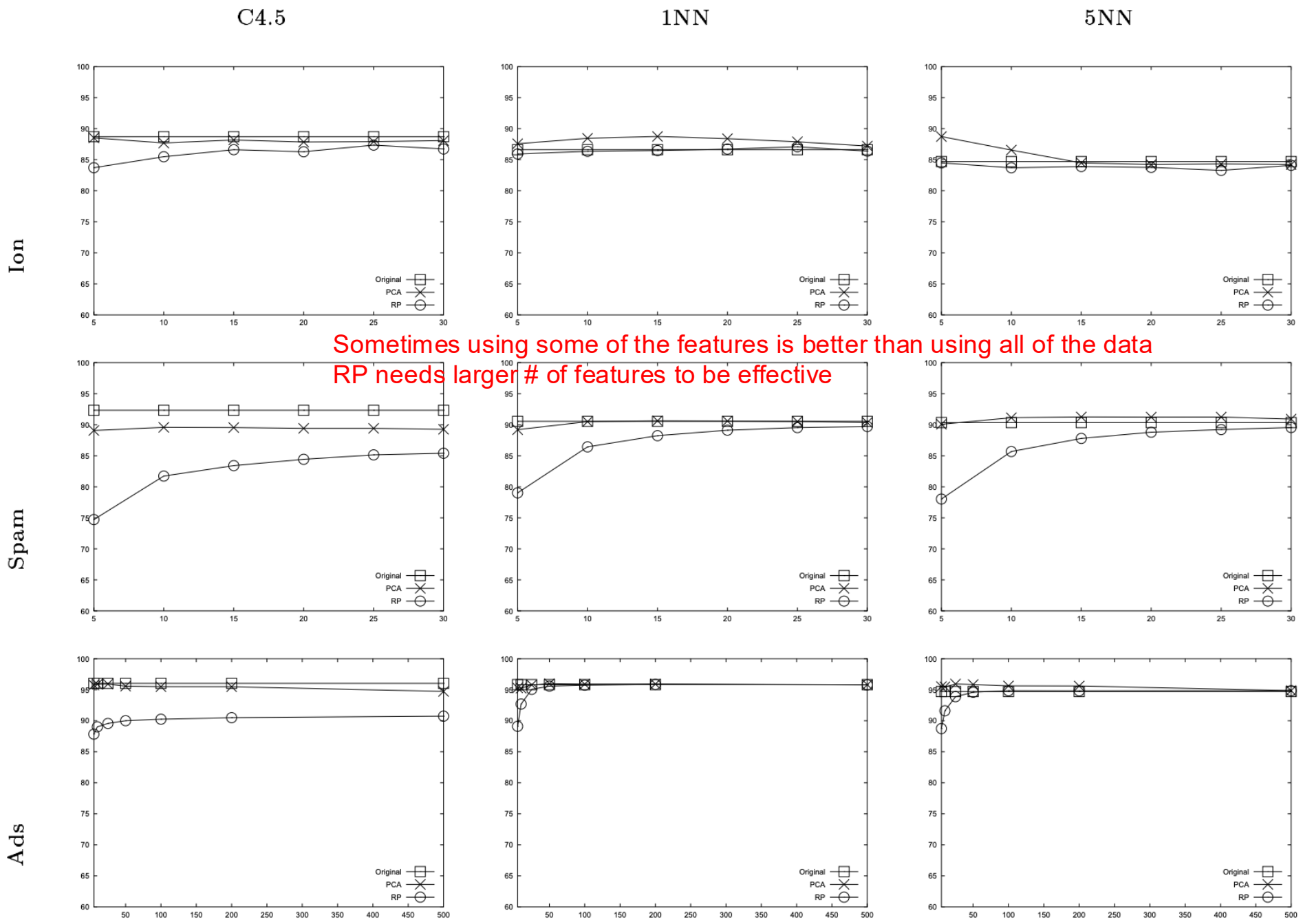
# Workflow for dimensionality reduction in ML

1. Reduce the number of features by projecting into lower dimensions (need to pick #of dimensions via cross validation)
2. Build a model using your favorite technique
3. ???
4. Profit

**Table 1: Description of Datasets**

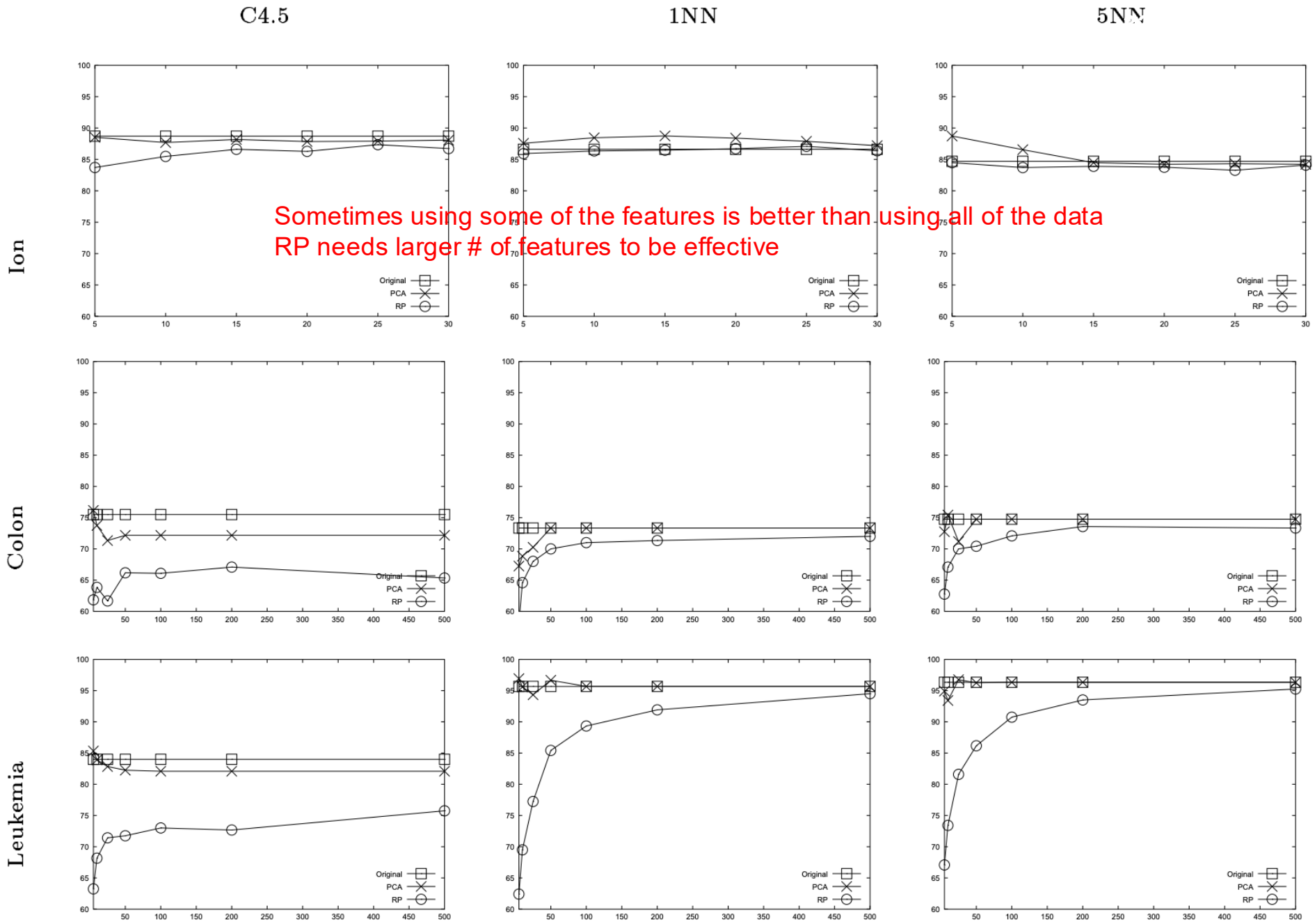
| Name | # Instances | # Attributes |
|---|---|---|
| Ionosphere | 351 | 34 |
| Colon | 62 | 2000 |
| Leukemia | 72 | 3571 |
| Spam | 4601 | 57 |
| Ads | 3279 | 1554 |

Experiments with Random Projections for Machine Learning
http://archive.dimacs.rutgers.edu/Research/MMS/PAPERS/rp.pdf




Sometimes using some of the features is better than using all of the data

RP needs larger # of features to be effective




Sometimes using some of the features is better than using all of the data
RP needs larger # of features to be effective

# Visualization

Methods covered

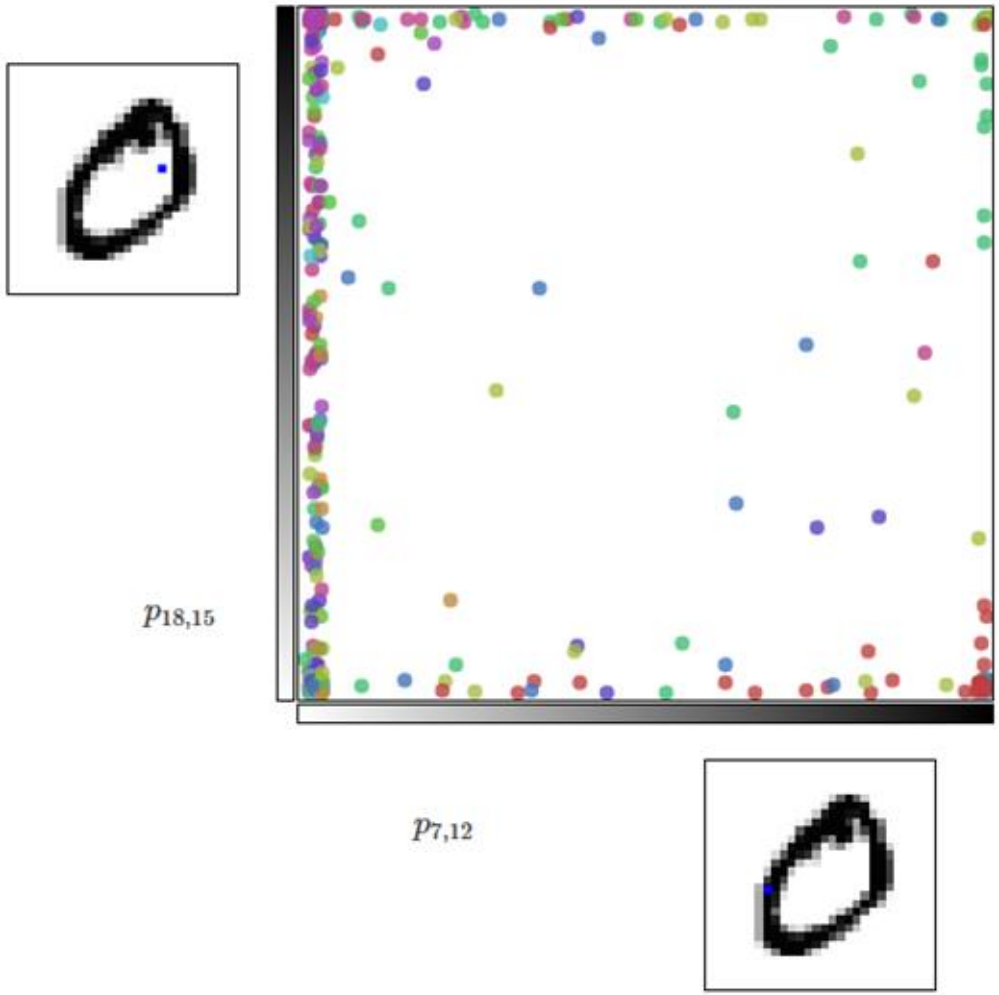
1. Reducing reconstruction error: PCA
2. Keeping direction of maximum separability: LDA
3. Preserving distance (globally): RP

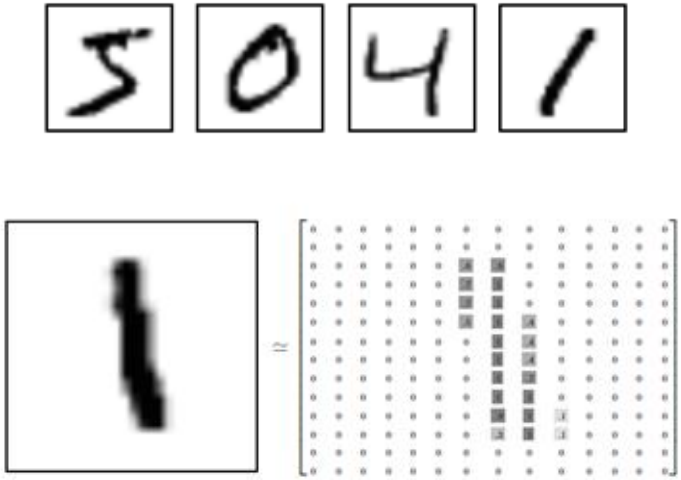
These are usually useful for downstream machine learning methods. (Classification/Regression/Clustering)

But what if we, as humans, want to get a sense of our data?

1. Interpretability (in some sense): ????
   - Check if our dimensionality reduction is good
   - Analyze clusters from k-mean, GMM, etc.

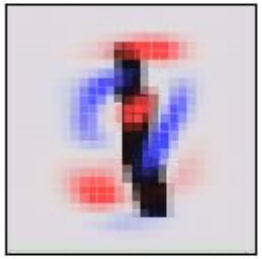
# Visualizing MNIST

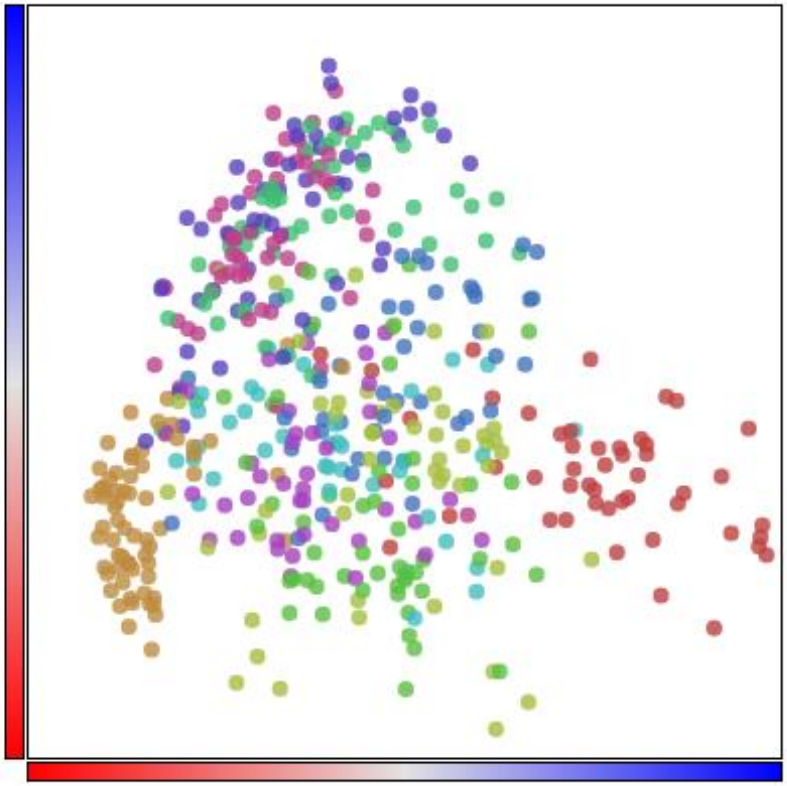


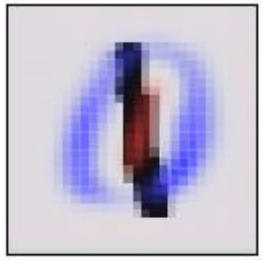



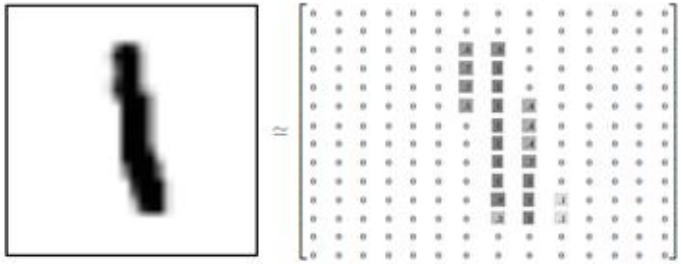
http://colah.github.io/posts/2014-10-Visualizing-MNIST/

# PCA with MNIST









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# t-distributed Stochastic Neighbor Embedding (t-SNE)

Preserves neighbor (preserves local distance).

- Things close together should be close together in the projected space
- Prefer using few projected dimensions (2-3)

# Defining neighbors

Define $P_{j|i}$ probability that *i* would pick *j* as its neighbor

Assume *i* picks proportional to Gaussian centered at *i*

$$p_{j|i} = \frac{exp(-||x_i - x_j||^2)/2\sigma_i^2}{\sum_{k \neq i} exp(-||x_i - x_k||^2)/2\sigma_i^2}$$

$P_{i|i}$ = 0 since we don't want to have it pick itself

The variance is fixed to some value.

# Defining neighbors

Define $p_{j|i}$ probability that *i* would pick *j* as its neighbor

Assume *i* picks proportional to Gaussian centered at *i*

$$p_{j|i} = \frac{exp(-||x_i - x_j||^2)/2\sigma_i^2}{\sum_{k \neq i} exp(-||x_i - x_k||^2)/2\sigma_i^2}$$

When projected to set of points $\{y_i\}$, define $q_{j|i}$ the probability that *i* would pick *j* in embedding/latent space

$$q_{j|i} = \frac{exp(-||y_i - y_j||^2)}{\sum_{k \neq i} exp(-||y_i - y_k||^2)}$$

We set the variance in the y space to be 1/sqrt(2)

# Defining neighbors

$$p_{j|i} = \frac{exp(-||x_i - x_j||^2)/2\sigma_i^2}{\sum_{k \neq i} exp(-||x_i - x_k||^2)/2\sigma_i^2}$$

$$q_{j|i} = \frac{exp(-||y_i - y_j||^2)}{\sum_{k \neq i} exp(-||y_i - y_k||^2)}$$

We expect p and q to be the same -> small distance

How to measure distance between probability functions?

Kullback-Leibler (KL) divergence

# KL divergence

Distance between two distributions

$$D_{KL}(P||Q) = \sum_i P(i) log \frac{P(i)}{Q(i)} = -\sum_i P(i) log \frac{Q(i)}{P(i)}$$

Note $D_{KL}(P||Q) \neq D_{KL}(Q||P)$ (Not a real distance)

Always positive. Equals 0 iff Q = P at every point.

P(head) = 0.5 P(tail) = 0.5

Q(head) = 0.7 Q(tail) = 0.3

$D_{KL}(P||Q)$ = 0.5*ln 0.5/0.7 + 0.5*ln 0.5/0.3 = 0.087

$D_{KL}(Q||P)$ = 0.7*ln 0.7/0.5 + 0.3*ln 0.3/0.5 = 0.082

# Loss function

$$p_{j|i} = \frac{exp(-||x_i - x_j||^2)/2\sigma_i^2}{\sum_{k \neq i} exp(-||x_i - x_k||^2)/2\sigma_i^2} \qquad q_{j|i} = \frac{exp(-||y_i - y_j||^2)}{\sum_{k \neq i} exp(-||y_i - y_k||^2)}$$

We expect p and q to be the same -> small distance

Loss function

All points i

KL computes over j

$$\sum_i D_{KL}(p_i || q_i)$$

$$D_{KL}(P||Q) = \sum_j \boxed{P(j)} log \frac{P(j)}{Q(j)}$$

Note P can be considered as the weight for the distance

Where p is large but q is small -> large penalty

p is small but q is large -> small penalty

D(p||q) focuses on local structure in p

What are we minimizing wrt?

How to minimize loss?

# From SNE to t-SNE

Symmetric density

t-distributed

# Symmetric with joint probability

$$D_{KL}(P||Q) = \sum_i P(i) log \frac{P(i)}{Q(i)}$$

$p_{j|i} \neq p_{i|j}$

A point i far away from everything will have small $p_{j|i}$

- Location of points in q no longer matter

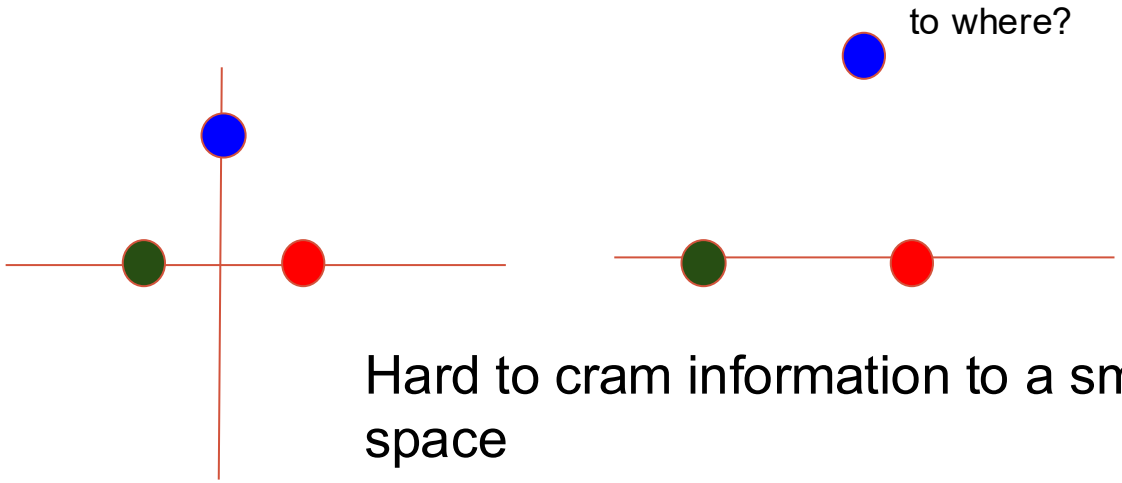
Create "joint probability" $p_{ij} = p_{ji} = (p_{i|j} + p_{j|i})/2$

- Each data point will contribute to the loss

Use instead of conditional probability in KL divergence

# t-distributed

- “Crowding problem”
- In N dimension, you can have N+1 points at equal distance. But you cannot model this in smaller dimension



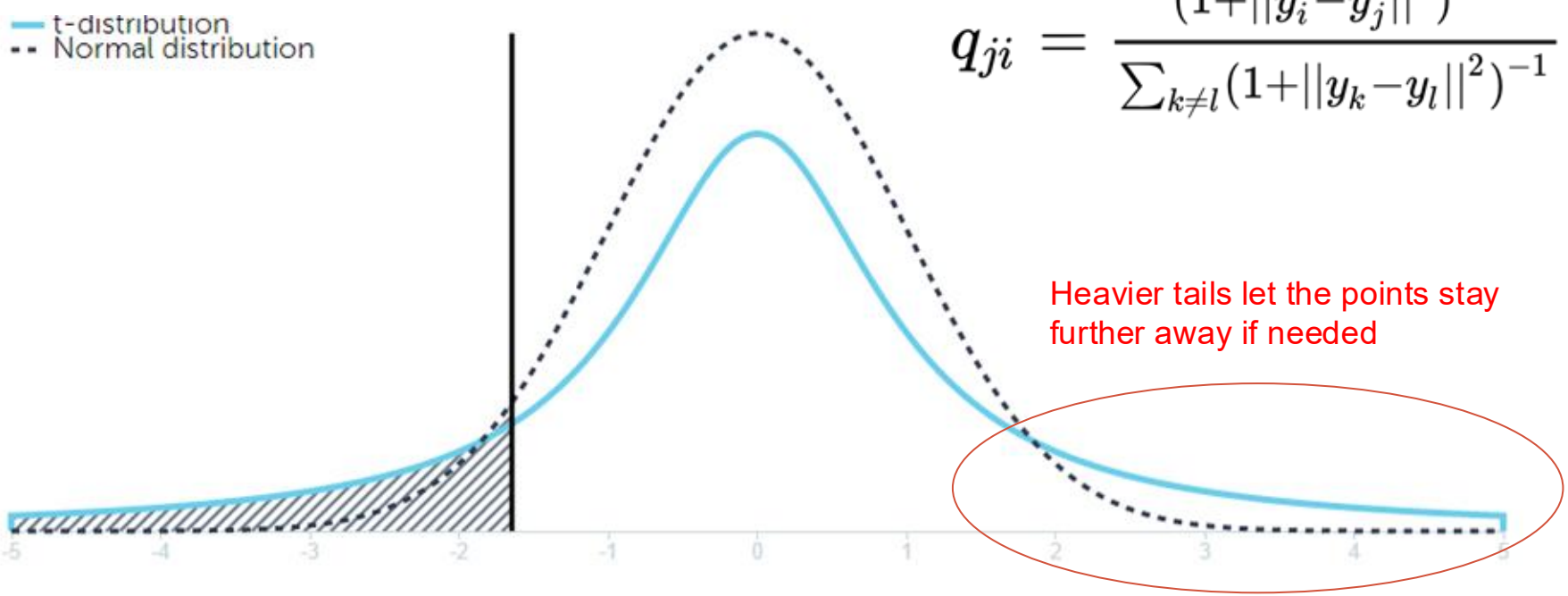

Hard to cram information to a smaller dimension space
Points tend to cram in the embedding space (no gap)

# t-distributed

- Instead of Gaussian for q we use student's t distribution

$$q_{ji} \propto (1 + ||y_i - y_j||^2)^{-1}$$

$$q_{ji} = \frac{(1+||y_i-y_j||^2)^{-1}}{\sum_{k \neq l}(1+||y_k-y_l||^2)^{-1}}$$

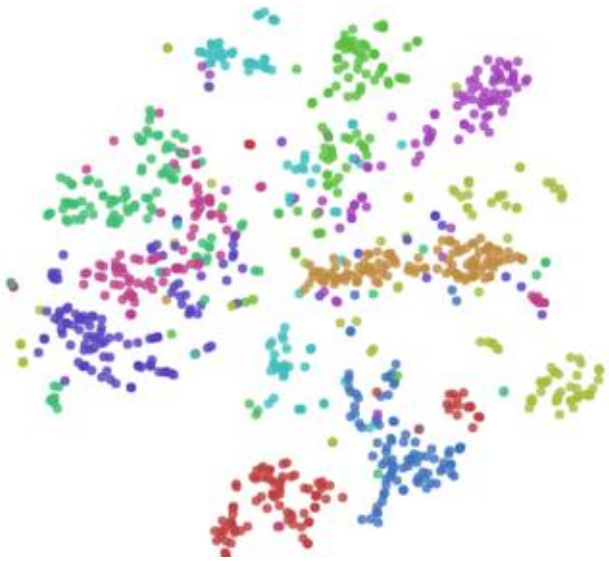



Heavier tails let the points stay further away if needed

http://rpsychologist.com/d3/tdist/

# Crowding and t-SNE

Sammon's mapping
(crowding problem)



t-SNE

http://colah.github.io/posts/2014-10-Visualizing-MNIST/

# Variance

$$p_{j|i} = \frac{exp(-||x_i - x_j||^2)/2\sigma_i^2}{\sum_{k \neq i} exp(-||x_i - x_k||^2)/2\sigma_i^2}$$

How to set the variance of our original space?

A single variance for all points is not ideal.

- Want small variance for dense parts
- Want big variance for sparse parts

Set variance by amount of neighbors you want!

How to quantify amount of neighbors?

# Perplexity

$$p_{j|i} = \frac{exp(-||x_i - x_j||^2)/2\sigma_i^2}{\sum_{k \neq i} exp(-||x_i - x_k||^2)/2\sigma_i^2}$$

$$Perp(P_i) = 2^{H(P_i)}$$

$$H(P_i) = -\sum_j p_{j|i} log_2 p_{j|i}$$ Entropy

Perplexity of $P_i$ represents effective amount of neighbors for the point i

Set Perp($P_i$) then t-SNE algorithm searches for the corresponding variance

Typical values for perplexity 5 to 50

# t-SNE summary

Goal: preserves local neighbors

Gradient-based -> need multiple runs to see the best

Two parameters: #iteration, perplexity

Does not learn a projection (unlike PCA, LDA, RP)

- If you have a new sample, you have to re-run the whole thing
- Used for visualization only

# UMAP: Uniform Manifold Approximation and Projection

t-SNE loses structure of important feature dimension or structure (global vs local structure)

Draw upon “inspiration” from Manifold learning theory

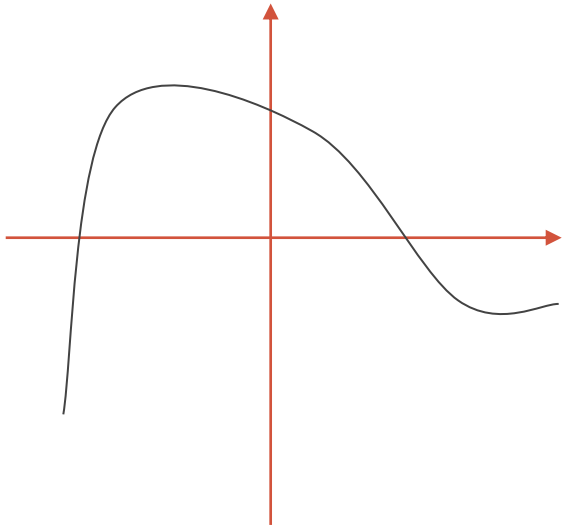
# UMAP big picture

1. Define a new kind of probability for being neighbor
   a. Based on distance to neighbors
2. Do SGD just like t-SNE

# Manifold

Manifold: a non-linear space but locally linear

Data can be represented in 2-d but lies in 1-d manifold



UMAP defines a manifold based on the neighbors of points

# UMAP's distance function

Given fixed k (number of neighbors)

Let $X_1, \ldots, X_k$ be the k-nearest neighbors of X

The distance between $X_i$ and $X_j$ is

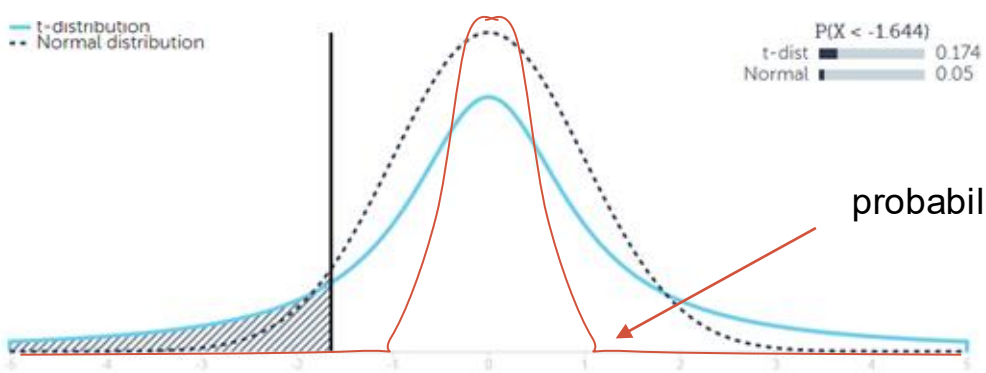
$$w_i(X_i, X_j) = \exp(-(d(X_i, X_j) - \rho_i)/\sigma_i)$$

diameter of the neighbors so it maxes at 1

Think of this w as giving "probability" that two points are connected

L2 distance in original space

distance to nearest neighbor so it starts from 0




probability becomes 0 after the $k_{th}$ neighbor

# UMAP's distance function

Given fixed k (number of neighbors)

Let $X_1, \ldots, X_k$ be the k-nearest neighbors of X

The distance between $X_i$ and $X_j$ is

$$w_i(X_i, X_j) = \exp(-(d(X_i, X_j) - \rho_i)/\sigma_i)$$

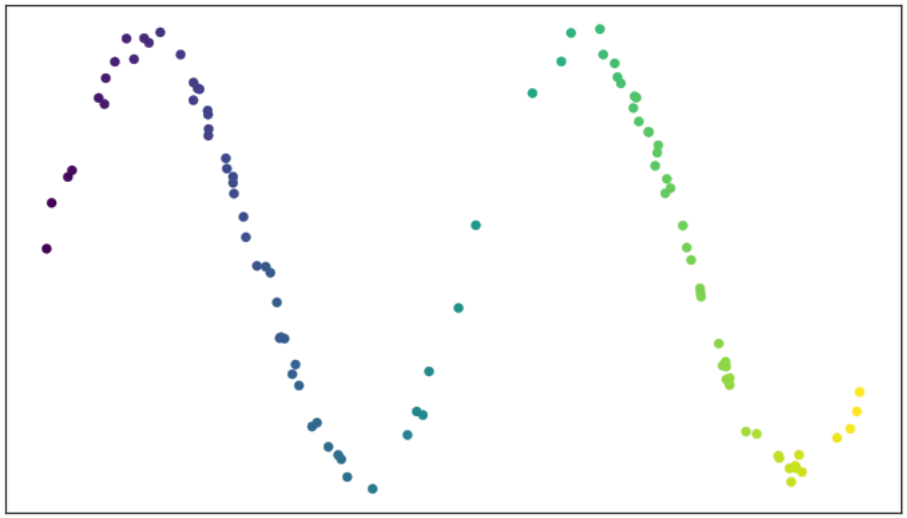
diameter of the neighbors so it maxes at 1

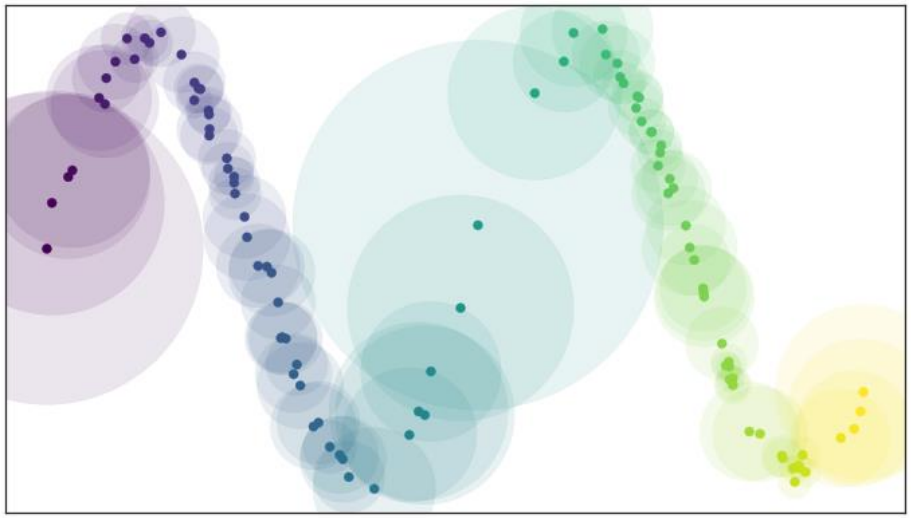
L2 distance in original space

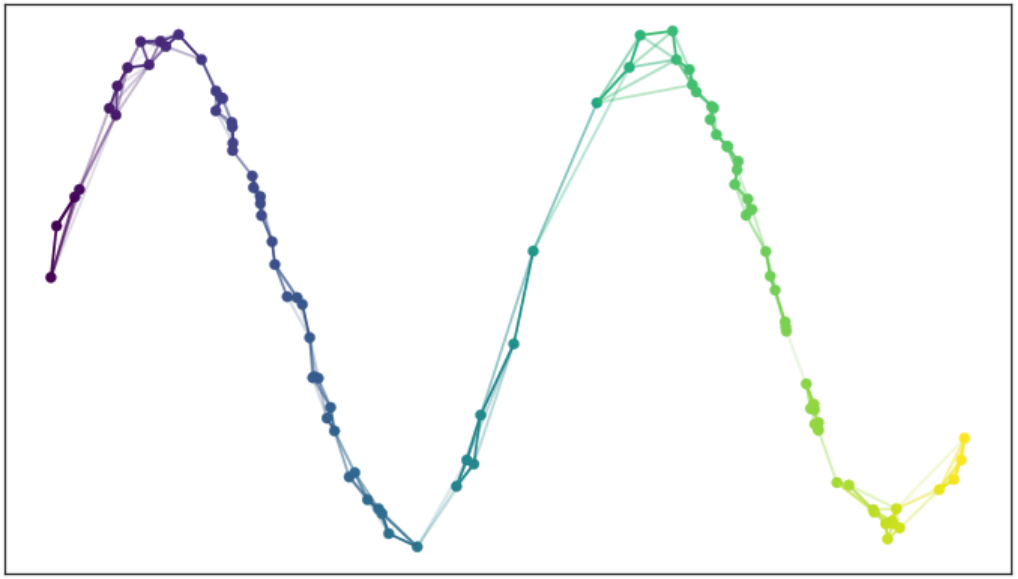
distance to nearest neighbor so it starts from 0

Make the distance symmetric by

$$w(X_i, X_j) = w_i(X_i, X_j) + w_j(X_j, X_i) - w_i(X_i, X_j)w_j(X_j, X_i)$$







https://umap-learn.readthedocs.io/en/latest/how_umap_works.html

# UMAP optimization

From the weights minimize the cross-entropy between original and projected points (in terms of w)

Distance in original space

$$\sum_{e \in E} w_h(e) \log\left(\frac{w_h(e)}{w_l(e)}\right) + (1 - w_h(e)) \log\left(\frac{1 - w_h(e)}{1 - w_l(e)}\right)$$

All neighbors

Distance in new space

Similar to KL metric in t-SNE

Loss function

All points i

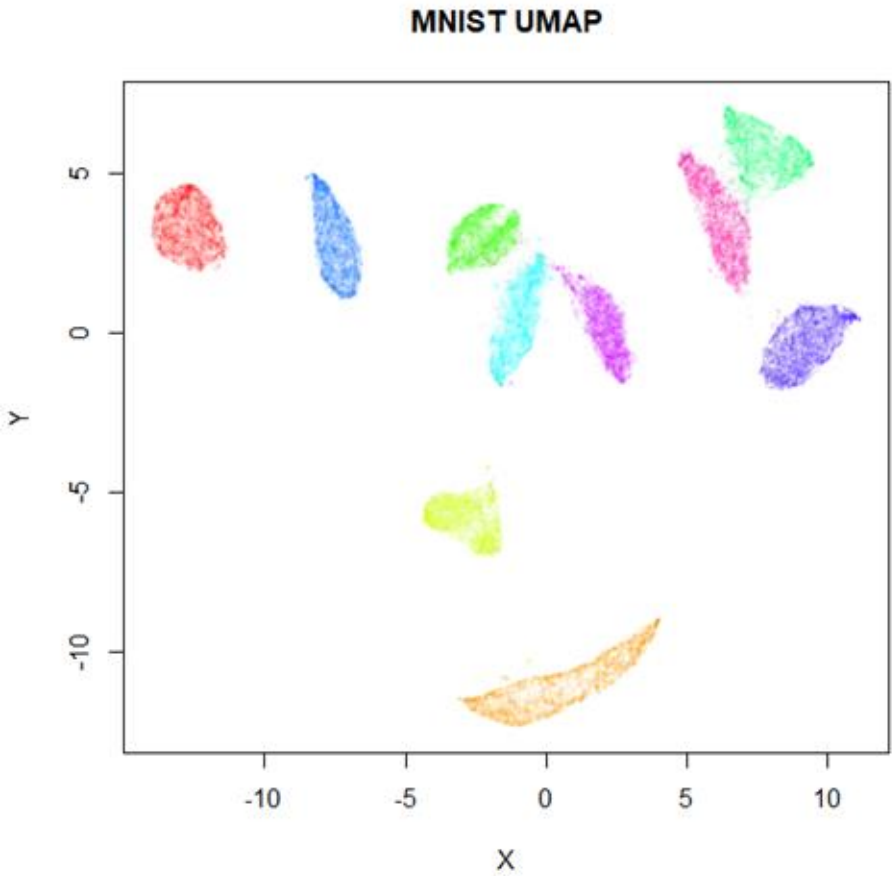
KL computes over j

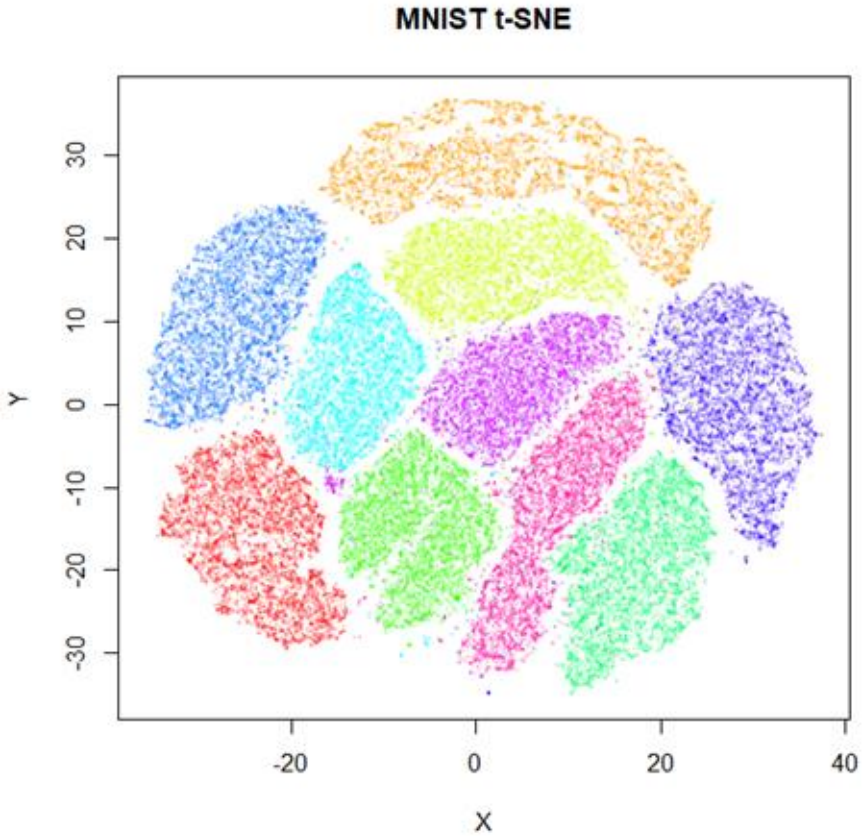
$$\sum_i D_{KL}(p_i || q_i)$$

$$D_{KL}(P||Q) = \sum_j \boxed{P(j)} log \frac{P(j)}{Q(j)}$$

Use gradient descent to optimize

# MNIST



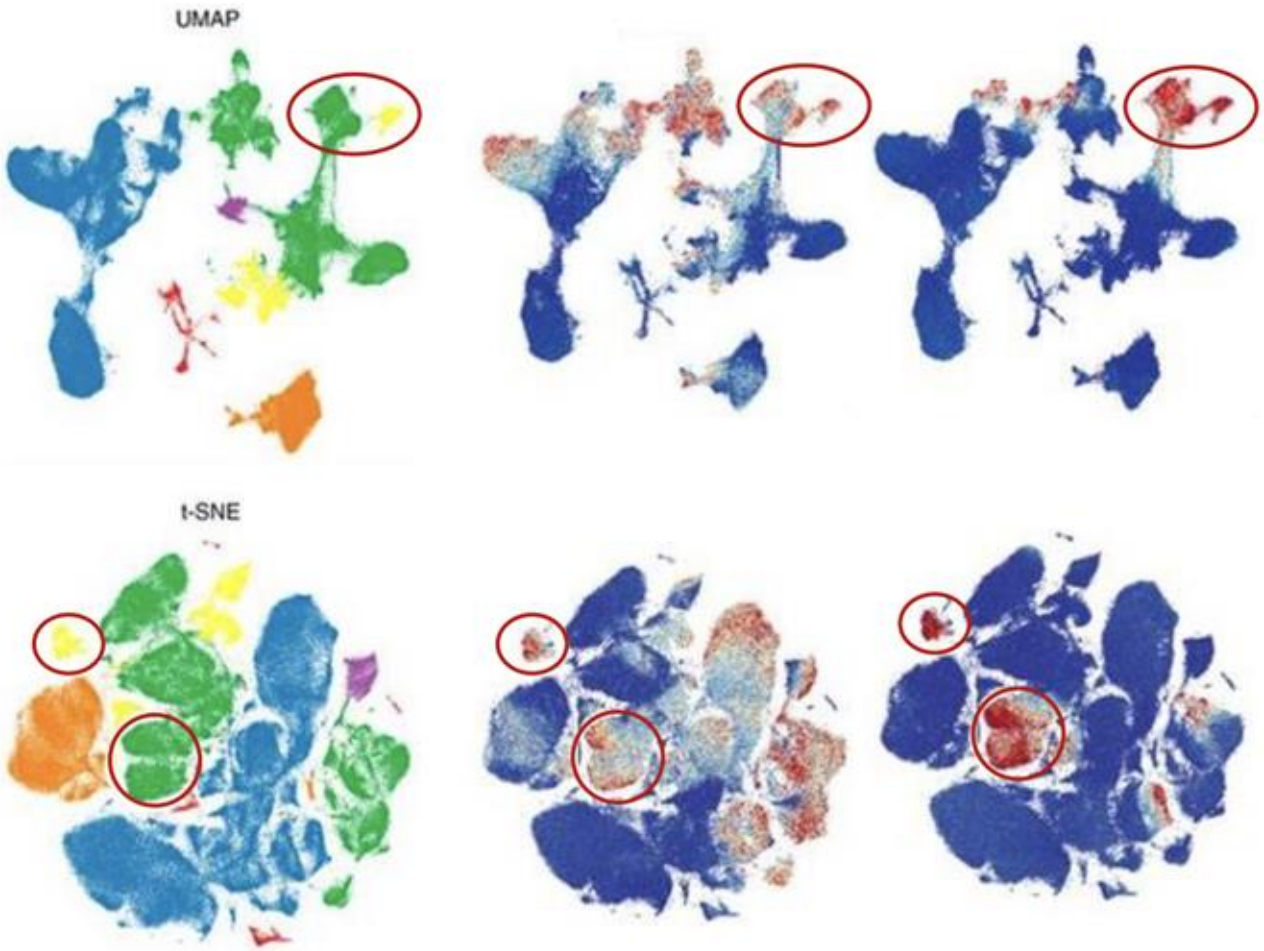




https://jlmelville.github.io/uwot/umap-examples.html

# Dimensionality reduction for visualizing single-cell data using UMAP

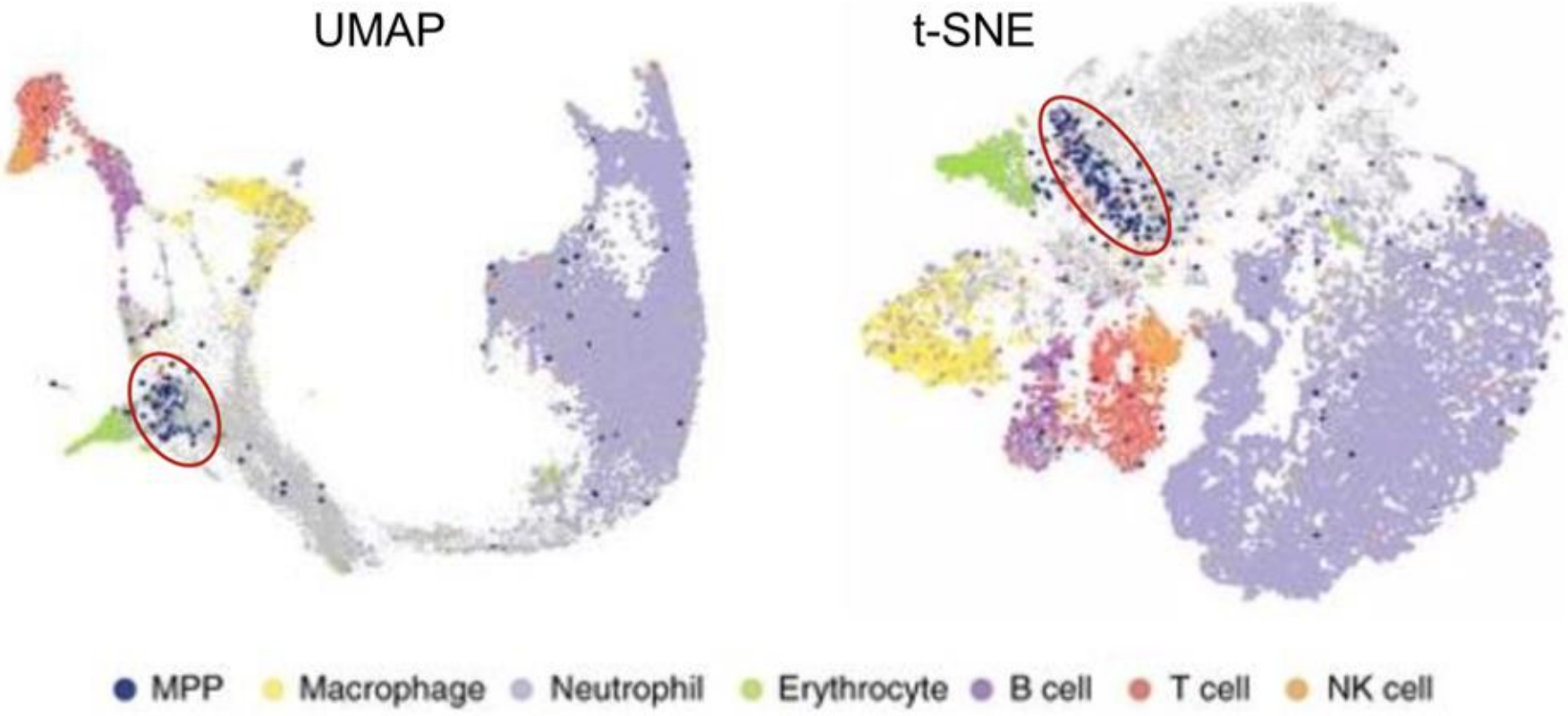
Analysis | Published: 03 December 2018

nature biotechnology

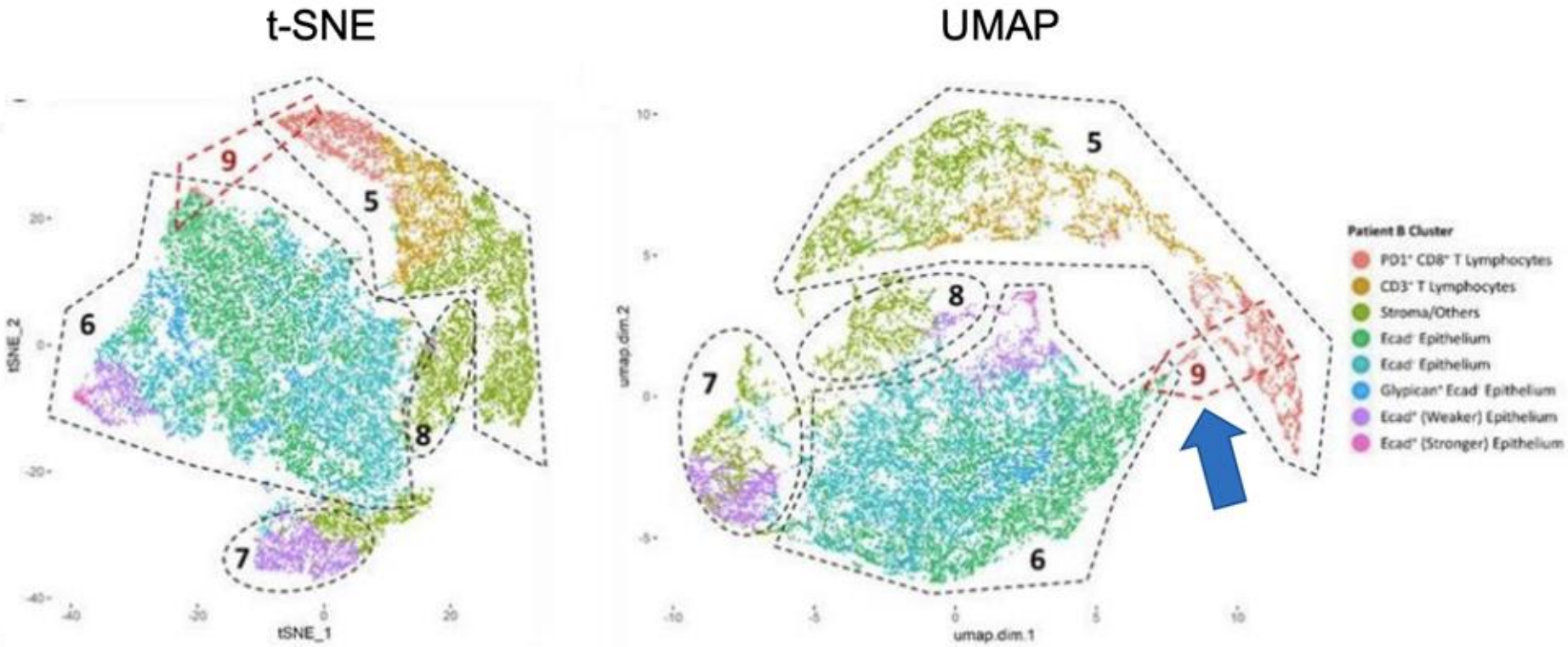



- UMAP can capture cross-class relationship

# UMAP vs t-SNE on continuous classes

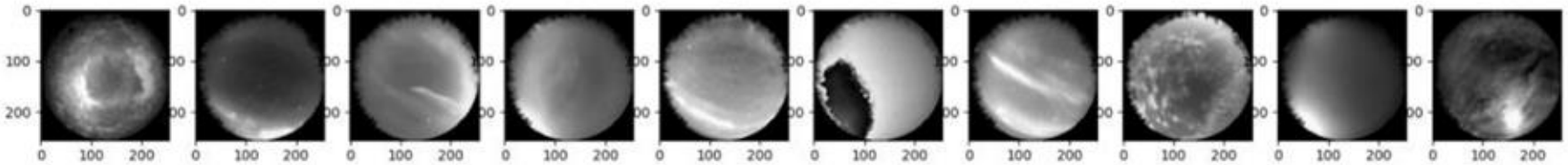


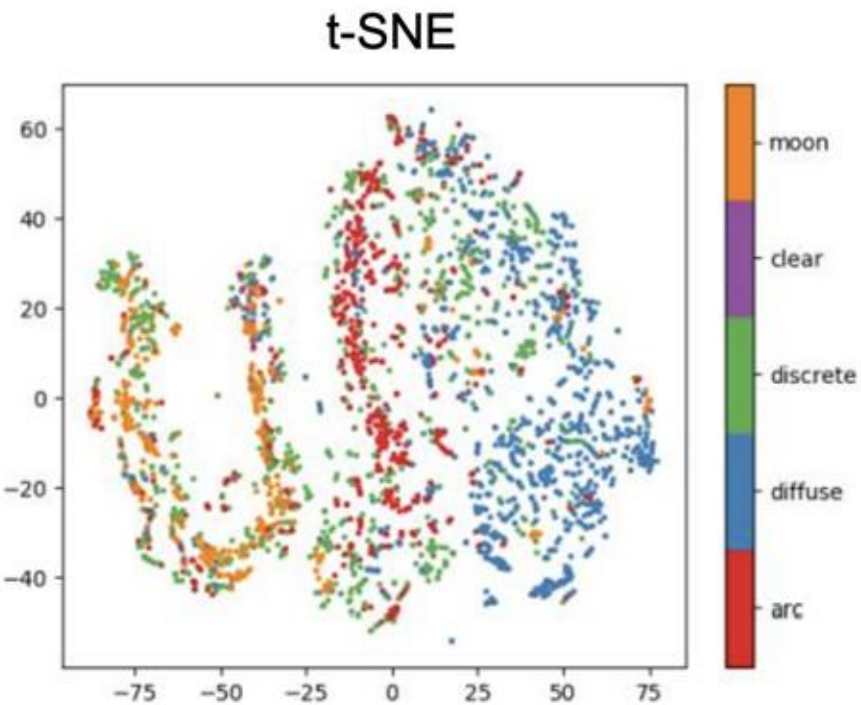
# UMAP find continuous transition across classes

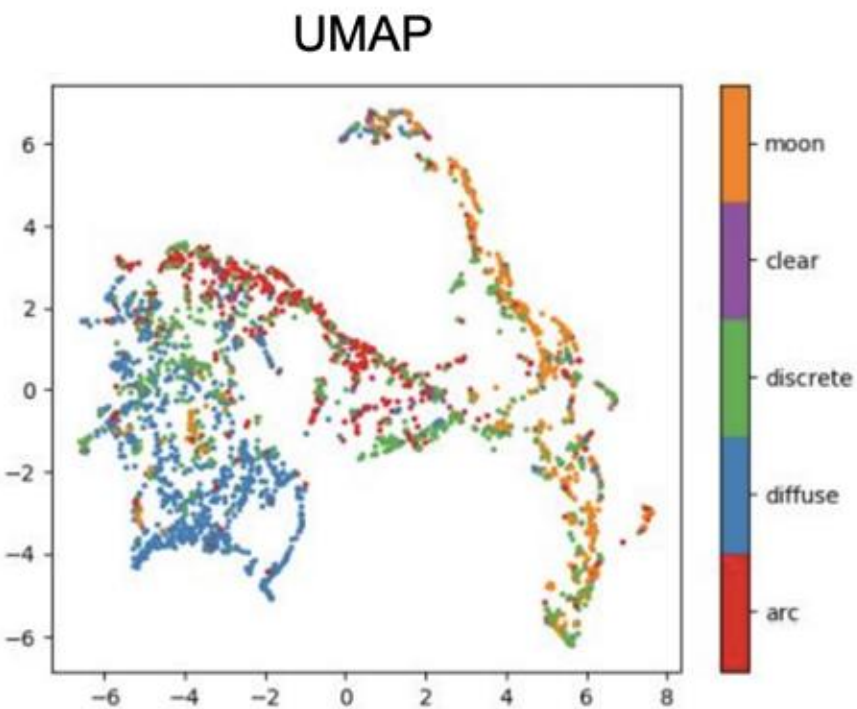


# Correlation of Auroral Dynamics and GNSS Scintillation with an Autoencoder

33rd Conference on Neural Information Processing Systems (NeurIPS 2019), Vancouver, Canada.

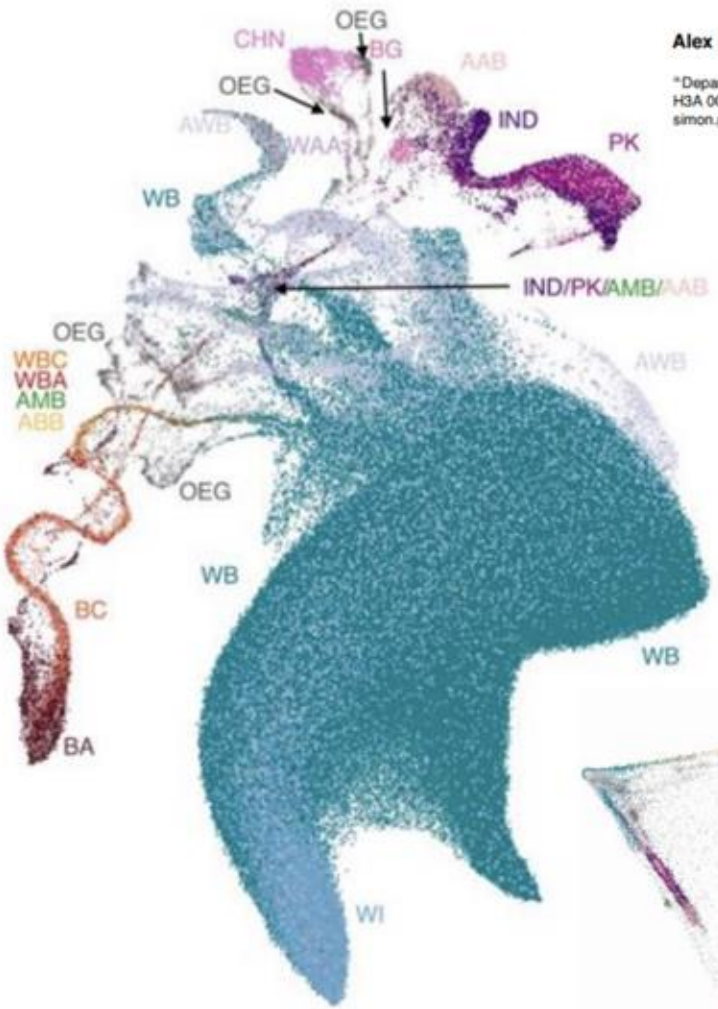




# UMAP vs t-SNE on population genomics

UMAP

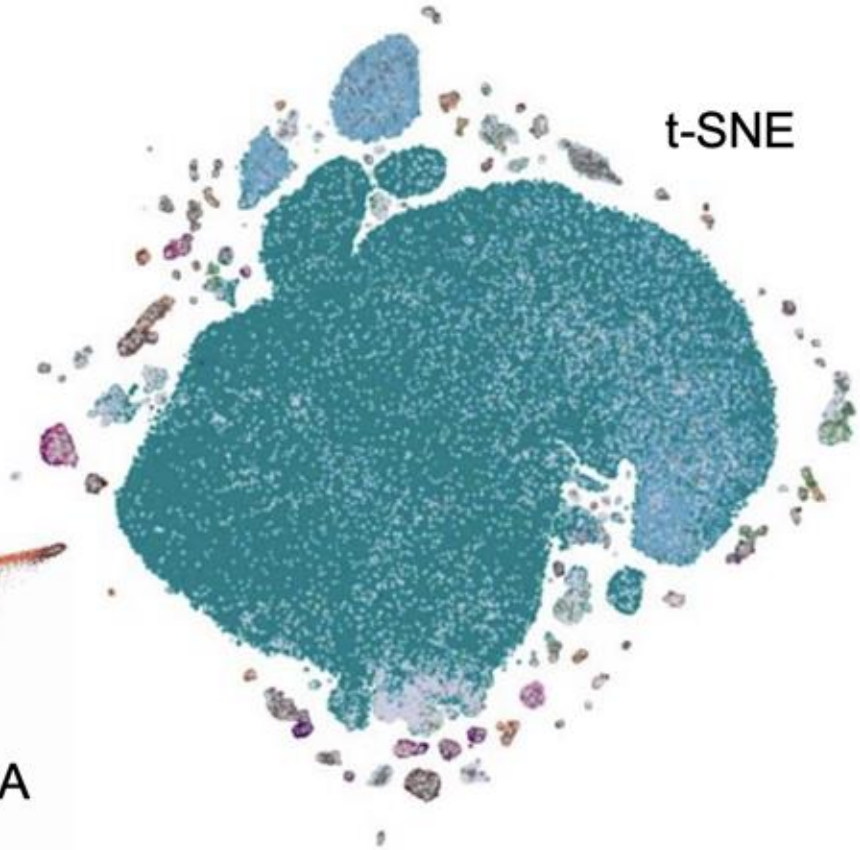



## Revealing multi-scale population structure in large cohorts

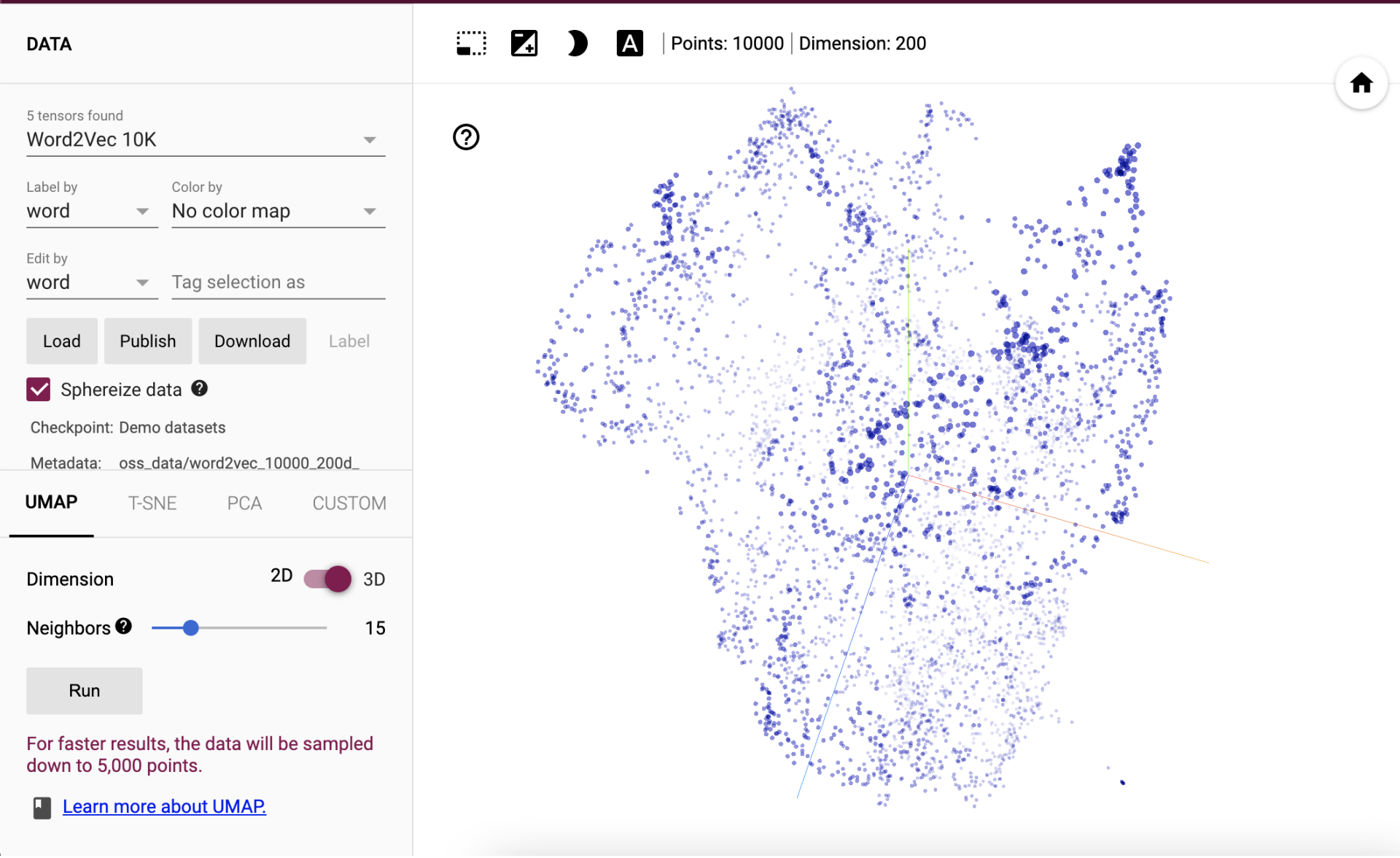
**Alex Diaz-Papkovich**[a,b], **Luke Anderson-Trocmé**[b,c], **and Simon Gravel**[b,c,1]

[a]Department of Quantitative Life Sciences, McGill University, Montreal, QC, H3A 0G1 Canada; [b]McGill University and Genome Quebec Innovation Centre, Montreal, QC, H3A 0G1, Canada; [c]Department of Human Genetics, McGill University, Montreal, QC, H3A 0G1, Canada. [1]To whom correspondence should be addressed. E-mail: simon.gravel@mcgill.ca

PCA



t-SNE

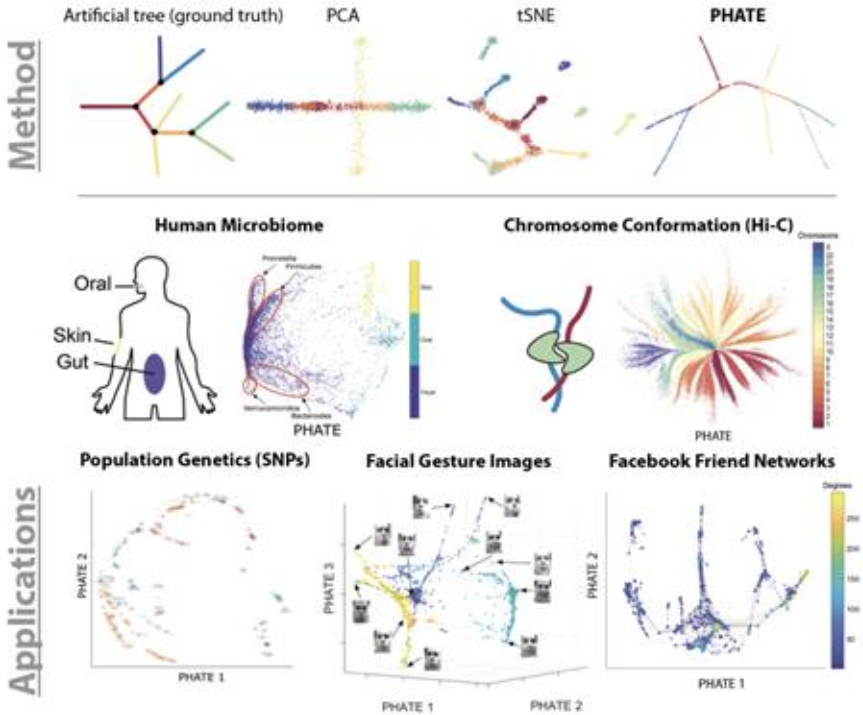



https://projector.tensorflow.org/

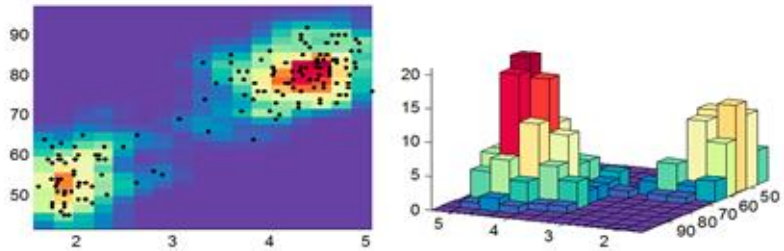
# Other improvements

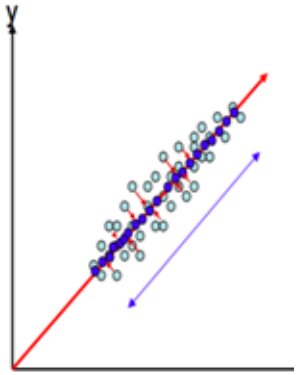
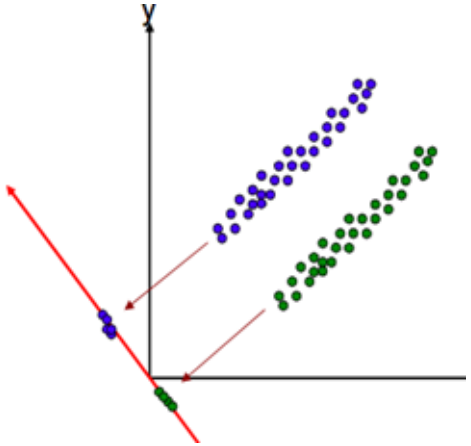
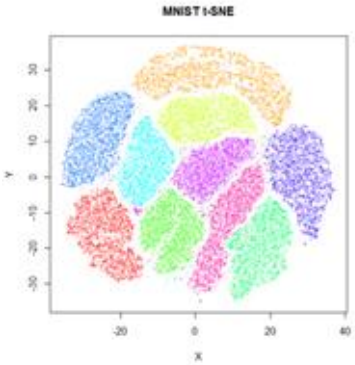
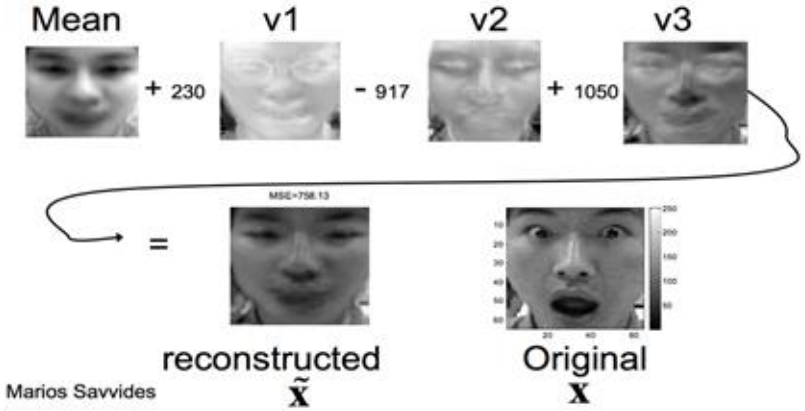
PHATE (https://www.krishnaswamylab.org/projects/phate)

Learn local structure/distance (just like UMAP) and global structures (using diffusion maps to traverse the graph). Probably better, but not popular yet.



# Summary




- Curse of dimensionality
- PCA
- LDA
  - PCA+LDA
- Random projection
- tSNE
- Homework